

NUMBERS

1.1 ROMAN NUMBERS

Read and write roman numbers up to 20

We have learnt numbers up to 1000 which are Hindi Arabic numbers. Roman numbers were used in ancient Rome. These are still in use in different ways.



Roman numbers can be seen on the dials of watches and clock towers in big cities. Sometimes these numbers are also used in books and examination papers.

Roman numbers from 1 to 20 are based on the following symbols.

Value	Symbol
1	I
5	V
10	X

Read Roman numbers from 1 to 20.

Arabic Numbers	Roman Numbers		Hindi Arabic Numbers	Roman Numbers	
	Capital	Small		Capital	Small
1	I	i	11	XI	xi
2	II	ii	12	XII	xii
3	III	iii	13	XIII	xiii
4	IV	iv	14	XIV	xiv
5	V	v	15	XV	xv
6	VI	vi	16	XVI	xvi
7	VII	vii	17	XVII	xvii
8	VIII	viii	18	XVIII	xviii
9	IX	ix	19	XIX	xix
10	X	x	20	XX	xx

There is no Roman number to represent zero.

Teacher's Note

The teacher will draw the basic symbols for Roman numbers I, V, X on the board. The rules, as mentioned in the textbook, will be explained by writing them on board and constructing numbers accordingly. To check for understanding, the teacher will call for different children to come to the board and write a given number in Roman numerals. Teacher should develop Roman numbers from card. He/she should also make groups of students and engage them in making Roman numbers equivalent to given number up to 20.

Unit 1**NUMBERS****(Roman Numbers)****Rules for writing Roman Numbers**

You can see that while we are writing 4 in Roman, we do not use IIII to represent 4.

Rule 1. The numerals I and X can be repeated up to three times only.

For example: 3 = III , 20 = XX

Rule 2. Writing a smaller numeral to the left of a Roman Numeral will make the number less than that much of its actual numerical value.

For example: 4 = IV , 9 = IX

Rule 3. Writing a smaller numeral to the right of a Roman numeral will make the number that much greater.

For example: 6 = VI , 7 = VII
8 = VIII , 15 = XV



Activity Complete:

$$8 = 5 + 1 + 1 + 1 = \text{VIII}$$

$$13 = \boxed{}$$

$$14 = \boxed{}$$

$$16 = \boxed{}$$

$$19 = \boxed{}$$

$$3 = \text{III}$$

$$4 = \boxed{}$$

$$6 = \boxed{}$$

$$8 = \boxed{}$$

$$12 = \boxed{}$$

Teacher's Note

Teacher will explain the rules for writings Roman numbers with more examples.

Unit 1**NUMBERS****(Roman Numbers)****EXERCISE 1**

1. Write down the following in Roman numbers.

5 _____, 7 _____, 11 _____, 9 _____

8 _____, 16 _____, 17 _____, 20 _____

2. Write the following in Hindi Arabic numbers.

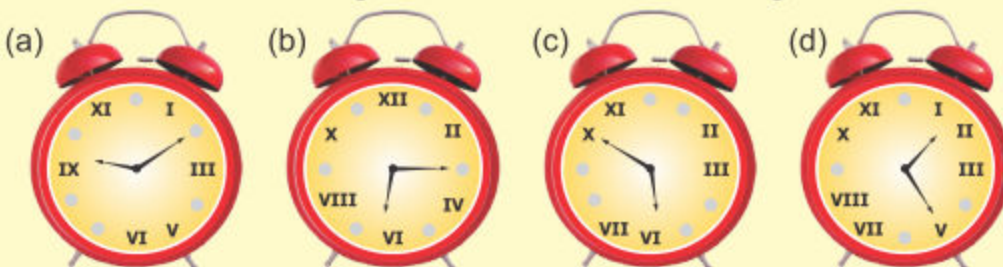
III _____, IV _____, VI _____, X _____

II _____, XV _____, XII _____, XIX _____

3. Write the missing numbers.

I		III			VI				X
		XIII				XVII			

4. Write the missing Roman numbers of the given clocks.



5. Match the following.

V	9	III	11
XII	2	X	8
IX	5	XI	3
VII	12	IV	10
II	7	VIII	4

An arrow points from the Roman numeral V in the first column to the Hindi Arabic number 5 in the second column.

6. Write Roman numbers from 1 to 20.

1.2 EVEN AND ODD NUMBERS

Starter:

The teacher will ask for 10 student volunteers to come to the front of the class.

- Each student has to find a partner (since there are 10 students, everyone will have a partner). Write 10 on the board under the word “even”.
- Explain to students that everyone has a partner because 10 is an “even” number.
- Ask 2 students for sitting down, 8 students standing. Everyone should have a partner. Write 8 on the board under “even”. Repeat this process with 6, 4, and 2 students.
- Ask students what they think would happen if there were 9 students standing. Would everyone have a partner? Repeat for 7, 5, 3, 1

Identify even and odd numbers up to 99 within a given sequence

EVEN NUMBERS

The numbers which are exactly divisible by 2 are even numbers.

Even numbers can produce pairs of objects with remainder zero.

ODD NUMBERS

The numbers which are not exactly divisible by 2 are odd numbers.

Odd numbers can not produce pairs of objects with zero remainder.

Unit 1**NUMBERS****(Even and Odd Numbers)**

Numbers	Objects	Even / Odd
1		Odd
2		Even
3		Odd
4		Even
5		Odd
6		Even
7		Odd
8		Even
9		Odd
10		Even

Teacher's Note

Teacher should help the students to identify even or odd numbers by using division by 2 and making pairs of the students and other objects in class.

Unit 1

NUMBERS

2, 4, 6, 8, 10, 12, 14, ... are even numbers.

Even numbers have 0, 2, 4, 6 or 8 at ones place.

1, 3, 5, 7, 9, 11, 13, 15, ... are odd numbers.

Odd numbers have 1, 3, 5, 7 or 9 at ones place.

I am odd;
if the digit at
ones place is:



I am even;
if the digit at
ones place is:



Write even or odd numbers within a given sequence

Look at the following chart:

0	1	2	3	4	5	6	7	8	9
10	11	12	13	14	15	16	17	18	19
20	21	22	23	24	25	26	27	28	29
30	31	32	33	34	35	36	37	38	39
40	41	42	43	44	45	46	47	48	49
50	51	52	53	54	55	56	57	58	59
60	61	62	63	64	65	66	67	68	69
70	71	72	73	74	75	76	77	78	79
80	81	82	83	84	85	86	87	88	89
90	91	92	93	94	95	96	97	98	99

The numbers given in red are odd numbers.

1, 3, 5, 7, 9, 11, ..., 99.

The numbers given in green are even numbers.

0, 2, 4, 6, 8, 10, 12, ..., 98.

Unit 1**NUMBERS****(Even and Odd Numbers)**

Example 1. Write even numbers between 3 and 11.

3, 4, 5, 6, 7, 8, 9, 10, 11

Thus, 4, 6, 8 and 10 are even numbers.

Example 2. Write odd numbers between 20 and 30.

20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30

Thus, 21, 23, 25, 27 and 29 are odd numbers.

Remember that

- Even number + 1 = Odd number
- Odd number + 1 = Even number

- Even + Even = Even
- Odd + Odd = Even
- Even + Odd = Odd

EXERCISE 2

1. Encircle (○) the even numbers from the following.

80, 83, 15, 98, 67, 78, 42, 27, 45

2. Encircle (○) the odd numbers from the following.

30, 38, 71, 32, 64, 69, 70, 11, 95

3. Write even numbers between 1 to 20.
4. Write odd numbers from 1 to 21.
5. After 30, write first five even numbers.
6. After 50, write first four odd numbers.
7. Write even numbers from 50 to 80.
8. Write odd numbers from 70 to 90.
9. Write (T) for the true and (F) for the false statement.

(i) 23 is an even number.	☐
(ii) 52 is an odd number.	☐
(iii) 15 is an odd number.	☐
(iv) 30 is an even number.	☐

1.3 PLACE VALUES

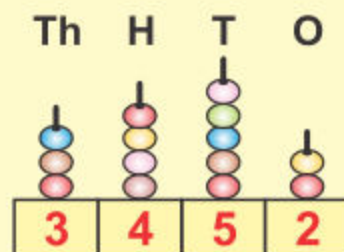
Identify the place values of numbers up to 6-digits

The 4-digit numbers are 1000, 1001, 1002, . . . , 9999.

Example 1: 3452 is a 4-digit number. The place value of each digit of number 3452 is as under.

$$\begin{aligned} 3452 &= 3 \text{ thousand} + 4 \text{ hundred} + 5 \text{ tens} + 2 \text{ ones} \\ &= 3000 + 400 + 50 + 2 \\ &= 3452 \end{aligned}$$

Th	H	T	O
3	4	5	2



Example 2: The place value of each digit in 654921 is as under

$$\begin{aligned} 654921 &= 6 \text{ hundred thousand} + 5 \text{ ten thousand} \\ &\quad + 4 \text{ thousand} + 9 \text{ hundred} + 2 \text{ tens} + 1 \text{ ones} \\ &= 600000 + 50000 + 4000 + 900 + 20 + 1 \\ &= 654921 \end{aligned}$$

H-Th	T-Th	Th	H	T	O
6	5	4	9	2	1

Unit 1

NUMBERS

(Place Values)



Activity

Write the place value of coloured digit in the following numbers

(i)

H	T	O
2	4	5

5 ones

(ii)

Th	H	T	O
8	0	6	2

(iii)

T-Th	Th	H	T	O
4	9	6	7	4

EXERCISE 3

1. Fill the boxes.

(i) $5806 = \square \text{ thousands} + \square \text{ hundreds} + \square \text{ tens} + \square \text{ ones}$

(ii) $51001 = \square \text{ ten thousands} + \square \text{ thousands} + \square \text{ hundreds} + \square \text{ tens} + \square \text{ ones}$

(iii) $12902 = \square \text{ ten thousands} + \square \text{ thousands} + \square \text{ hundreds} + \square \text{ tens} + \square \text{ ones}$

(iv) $442609 = \square \text{ hundred thousands} + \square \text{ ten thousands} + \square \text{ thousands} + \square \text{ hundreds} + \square \text{ tens} + \square \text{ ones}$

2. Identify the place value of coloured digit number given in boxes:

(i) $24 \boxed{3} 6$

(ii) $5 \boxed{4} 123$

(iii) $61237 \boxed{8}$

(iv) $\boxed{4} 129$

(v) $33 \boxed{5} 566$

(vi) $7 \boxed{2} 081$

(vii) $\boxed{9} 4123$

(viii) $567 \boxed{8} 12$

(ix) $81 \boxed{0} 703$

3. Complete the following:

No.	Number	H-Th	T-Th	Th	H	T	O
(i)	Six thousand four hundred forty two			6	4		2
(ii)	Twenty Five thousand six hundred thirty one		2			3	1
(iii)	Ninety two thousand two		9		0		2
(iv)	Five hundred twenty three thousand five hundred	5		3			0

Read and write given numbers up to 100,000 (hundred thousand) in numeral and in words

We have already learnt 1-digit, 2-digit and 3-digit numbers

4-digit number

Example:

Write 4235 in words.

Solution

Th	H	T	O
4	2	3	5

Four thousand two hundred thirty five



5-digit numbers

The number after 9999 is 10000
Read as "Ten Thousand"
i.e $9999 + 1 = 10000$

10000 is the smallest 5-digit number.



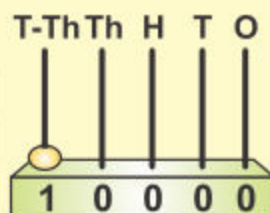
Unit 1

NUMBERS

(Place Value)

Place Value Chart for 10000

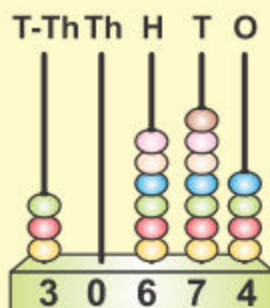
Ten Thousands	Thousand	Hundreds	Tens	Ones
T-Th	Th	H	T	O
1	0	0	0	0



Example: Write 30674 in words

Solution:

T-Th	Th	H	T	O
3	0	6	7	4



Thirty five thousand six hundred seventy four

6-digit numbers

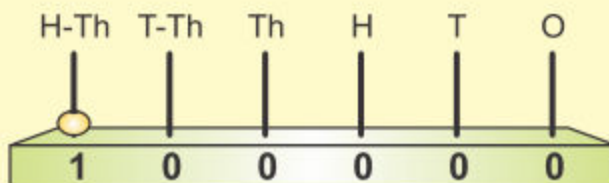
The number after 99999 is **100000**
 Read as "Hundred Thousand"
 i.e $99999 + 1 = 100,000$



100000 is the smallest 6-digit number

Place Value Chart for 100,000

Hundred Thousands	Ten Thousands	Thousand	Hundreds	Tens	Ones
H-Th	T-Th	Th	H	T	O
1	0	0	0	0	0



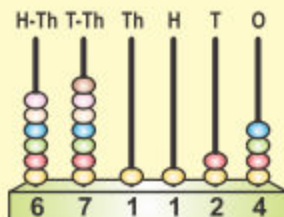
Teacher's Note

Teacher should do some more examples with the help of only place value chart.

Unit 1**NUMBERS****(Place Value)****Example:** Write 671124 in words**Solution:**

H-Th	T-Th	Th	H	T	O
6	7	1	1	2	4

Six hundred seventy eight thousand one hundred twenty four

**Look at the pattern.**

$$\begin{aligned}
 9 + 1 &= 10 \\
 99 + 1 &= 100 \\
 999 + 1 &= 1000 \\
 9999 + 1 &= 10000 \\
 99999 + 1 &= 100000
 \end{aligned}$$

100000 is the smallest 6-digit number.

**Activity 3****Fill in the blanks:**

- (1) The greatest 4-digit number is 9999.
- (2) The smallest 5-digit number is _____.
- (3) The greatest 5-digit number is _____.
- (4) The greatest 6-digit number is _____.

EXERCISE 4**1. Read and write each of the following numbers in words.**

- (i) 4156 (ii) 35678 (iii) 981254 (iv) 5060
 (v) 7120 (vi) 356782 (vii) 41524 (viii) 123456

2. Write the following in figures.

- (i) Eight thousand four hundred sixty one. _____
 (ii) Seventy two thousand six hundred ninety five. _____
 (iii) Fifty five thousand one. _____

3. Fill in the blanks:

- (i) Number before 951 is _____
 (ii) Number after 25645 is _____
 (iii) Number between 5420 and 5422 is _____

4. Continue the patterns.

(i) 2124 , 3124 , 4124 , _____ , _____ , _____

(ii) 51427 , 51437 , 51447 , _____ , _____ , _____

(iii) 123456, 223456, 323456 , _____ , _____ , _____

1.5 COMPARING AND ORDERING THE NUMBERS

Compare two numbers using symbols $<$, $>$ and $=$

Compare the number of the following objects:

How many candles
are in blue box?How many candles
are in yellow box?9 candles are more than 4 candles. We say **9 is greater than 4****Symbolically**, we write as **$9 > 4$** . The symbol " $>$ " stands for "is greater than"**Similarly**, 4 candles are less than 9 candles. We say **4 is less than 9****Symbolically**, we write as **$4 < 9$** . The symbol " $<$ " stands for "is less than"

Teacher's Note

The teacher will tell the class that they will be practicing comparing numbers. Ask for student volunteer to define the word compare. After some discussion, remind your class that compare means to see if one number is the same as, smaller than, or bigger than another number.

Write the terms greater than, less than, and equal to on the whiteboard. Prompt your class to tell you what each of these terms mean. After some students share what they believe each means, write the definition for each on the board. Remind the class that greater than means bigger, less than means smaller, equal to means the same.

Explain that as a class, everyone will review the symbols used to compare numbers

Teacher should help the students to compare numbers by using symbols ' $<$ ' and ' $>$ ' made from other materials/flash

**Activity**Use symbols $<$ or $>$.

(i) $5 > 3$

(ii) $20 \square 16$

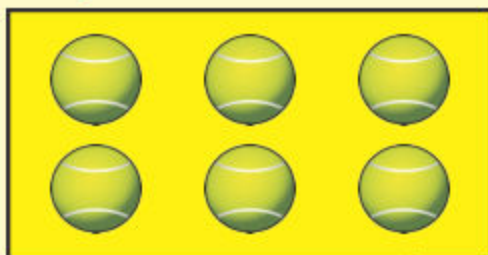
(iii) $36 \square 47$

(iv) $60 \square 80$

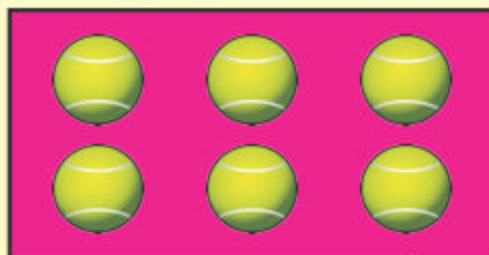
(v) $325 \square 300$

(vi) $960 \square 980$

Compare the number of the following objects:



How many balls are there?



How many balls are there?

Both boxes have equal number of balls. We say **6 is equal to 6**Symbolically, we write as $6 = 6$.

Now observe the following examples.

Example 1:Compare **245** and **6472**.**Solution:** 245 has 3 digits

6472 has 4 digits

So, $245 < 6472$ **Activity**Compare **98** and **123**.**Solution:** 98 has digits123 has digitsSo, 98 123**Example 2:**Compare **5423** and **967**.**Solution:** 5423 has 4-digits

967 has 3-digits

So, $5423 > 967$ **Activity**Compare **504** and **99**.**Solution:** 504 has digits99 has digitsSo, 504 99